



Frozen DINOv3 and Multilingual-E5 Embeddings with CatBoost for Multimodal Triage of Citizen Environmental Reports

Author Name^{1*}, Author Name²

¹Author Affiliation, Country

²Author Affiliation, Country

*Correspondence: E-mail: email@xx.yy

ABSTRACT

The manual triage process for environmental reports submitted by citizens remains a structural bottleneck within complaint-handling channels. Within Indonesia's e-government framework, the DKI Jakarta Provincial Government's Citizen Relationship Management (CRM) platform processed 185,852 reports in 2024 alone. Although the national SP4N-LAPOR! system records an average resolution time of 6.1 working days well within the 10-day target—field observations reveal that routing a single report to the appropriate agency can take nearly two hours due to manual reading and assessment stages. This study proposes and evaluates a multimodal pipeline workflow utilizing "early fusion" techniques to classify citizen reports—comprising images and Indonesian-language narratives—for routing to nine target agencies. By combining a frozen, self-supervised vision encoder (DINOv3-Large) with a multilingual text encoder (multilingual-E5-Large), the system generates L2-normalized 1024-dimensional embeddings, which are concatenated into a 2048-dimensional joint vector. Classification is subsequently performed using a CatBoost gradient-boosting head with Posterior Gaussian Sampling (PGS) virtual ensembles. The system is implemented using a FastAPI backend accessible via Android. The application directs users to the nearest agency by utilizing Haversine distance calculations. Paired significance tests and a replication of the pipeline on a later snapshot of the same portal, with per-sample predictions, show that PGS averaging

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lowers accuracy at a fixed checkpoint and slightly worsens calibration, whereas the plain maximum class probability of ordinary argmax inference, the mode served in production, is the stronger signal for deferring uncertain reports to human operators. These findings offer methodological and technical contributions to multimodal civic technology systems.

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Background and Problem Statement

Public reporting on urban environmental issues such as damaged roads, foul-smelling waste accumulation, fallen trees, traffic incidents, and vandalism has become a primary channel for citizen participation in smart city governance in Indonesia. However, in operational reality, these channels remain heavily reliant on manual review. For instance, the Jakarta CRM platform, managed by the DKI Jakarta Provincial Government, processed 185,852 citizen reports via thirteen integrated complaint channels in 2024, with over 95,000 of these reports submitted in the first half of the year alone (Jakarta Smart City, 2024a, 2024b). At the national level, the SP4N-LAPOR! System managed by the Ministry of State Apparatus Empowerment and Bureaucratic Reform has recorded 2.1 million reports involving 679 connected agencies, comprising 34 ministries, 101 institutions, and 544 local governments (Kementerian PANRB, 2024). This massive volume of reports necessitates a rapid and accurate triage process (sorting and prioritization) to ensure reports are forwarded to the responsible agencies without delay.

Empirical evidence indicates that manual triage processes rather than a lack of institutional capacity are the primary cause of delays. Reports from KemenPANRB itself show a national average follow-up time of 6.1 working days against a 10-day target as of early 2023 (Kementerian PANRB, 2023), while the Jakarta CRM platform records an average resolution time of approximately 110 hours (about 5 days) per report (General Affairs and Administration Bureau of the DKI Jakarta Provincial Secretariat, 2023). A field study found that the process of assigning a single report to the appropriate agency takes an average of nearly two hours, as sub-district-level staff must manually determine the correct recipient agency (Hanif et al., 2024). This workload increases linearly with the volume of reports on other regional platforms that receive thousands of reports daily. The sheer volume of data, combined with an imbalanced distribution of categories, poses a significant obstacle to processing, whether performed manually or automatically (Zahro et al., 2025).

In general, this premise has not been widely discussed in the literature on multimodal civic technology. However, the research is directly supported by prior studies most relevant to this field—specifically Hanif et al. (2026), who also utilized Jakarta CRM data. In their experiments involving an early-fusion combination—comprising a frozen DINOv2 image encoder, a frozen multilingual-E5 text encoder, and a CatBoost classification head calibrated via Posterior Gaussian Sampling (PGS). They achieved a weighted-F1 score of 80.73% across 53,885 reports spanning eight agencies. Evaluation results obtained through cross-validation strongly confirm that this general architectural family is highly suitable for the task. The present study is limited to the model level and does not implement the full pipeline, nor does it analyze the numerical or behavioral consequences of deploying PGS into a serving format;

furthermore, it evaluates only two encoder options per modality rather than mapping the entire available design space.

Previous research most relevant to this topic—covering other multimodal work related to citizen reporting—has largely focused on disaster imagery, document classification, or general-purpose English-language visual content (Radford et al., 2021; Praveen et al., 2023; Koshy and Elango, 2023). Meanwhile, recent civic technology prototypes for complaint routing typically rely on single-modality image classifiers built using lightweight tools like Teachable Machine (Shama et al., 2024) or employ joint embedding via CLIP without benchmarking against modern alternatives that are frozen and self-supervised (Tripathi et al., 2026). A related study utilizing only images from municipal e-government platforms also did not employ multimodal fusion; instead, it delegated text analysis to a separate model (Vrabie, 2025). This study extends the Hanif et al. (2026) lineage of work in four concrete directions: upgrading the visual backbone to DINOv3 and validating that choice through a full 4×4 encoder ablation (rather than a single pairing); scaling the dataset to 61,773 pairs across nine agencies; subjecting the PGS mechanism to a confound-controlled, statistically tested re-evaluation; and carrying the selected model through ONNX export validation, a production FastAPI/Flutter deployment, and real-user usability testing.

1.2 Research Gap and Objectives

This study addresses that gap by combining a frozen DINOv3-Large visual encoder and a frozen multilingual-E5-Large text encoder through early fusion, followed by a CatBoost (Prokhorenkova et al., 2018) classification head that uses Posterior Gaussian Sampling (Ustimenko et al., 2023), applied to an authentic dataset scraped from the Jakarta CRM public complaint portal. The dataset pairs images and Indonesian-language narratives, consolidated from 487 raw agency (SKPD) labels into nine operational target classes.

2. Materials and Methods

The study follows the Design Science Research Methodology (DSRM; Peffers et al., 2007), whose six stages — problem identification and motivation, objective definition, design and development, demonstration, evaluation, and communication — were selected because the principal research output is a system artefact: a multimodal classification pipeline, an inference server, and a client application intended to resolve the manual-triage problem described in Section 1. Figure 1 summarizes the overall research framework, linking the problem, the identified gap, the five research questions, the method, the empirical evidence gathered for each question, and the resulting conclusion.

2.1 Dataset Acquisition and Preparation

The dataset originates from the Jakarta CRM public-complaint portal, obtained via web scraping. Each raw record combines an attached photograph, a free-text Indonesian narrative, and a raw agency (SKPD) label. From an initial 88,302 scraped reports carrying 487 distinct raw SKPD names, the preparation pipeline removed approximately 30% duplicate narrative rows (copy-paste submissions), discarded rows with unreadable or corrupted images, and consolidated the 487 raw labels into nine operational target agencies: Dinas Bina Marga, Satuan Polisi Pamong Praja, Dinas Perhubungan, Kelurahan, Dinas Pertamanan dan Hutan, Dinas Sumber Daya Air, Dinas Cipta Karya/Tata Ruang/Pertanahan, Badan Pembinaan BUMD, and a residual “Other agencies” catch-all class. This yielded 61,773 clean, paired image-text samples. Because image and text originate from the same report row, the data

was split once at the row level — not independently per modality using a stratified shuffle split (seed = 42, ratio 70:15:15), keeping inter-partition class-proportion deviation below 0.01% and guaranteeing that no row appears in more than one partition. This produced 43,241 training, 9,266 validation, and 9,266 test pairs.

2.2 Multimodal Preprocessing, Frozen Encoders, and Early Fusion

Figure 2 details the end-to-end multimodal pipeline, from the raw CRM record to the deployed inference server. The image and text pathways are processed independently and merged only at the fusion stage. Images are resized to a square 224×224 without center-cropping (following the DINOv3 AutoImageProcessor default), normalized per channel with ImageNet statistics, and converted to NCHW float32 tensors before being passed through DINOv3-Large in evaluation (frozen) mode to yield a 1024-dimensional embedding. Text narratives are cleaned via regex-based URL removal and whitespace collapsing, prefixed with the mE5-Instruct instruction template, tokenized with SentencePiece (max length 256 tokens, covering roughly 98.5% of narratives without truncation), and mean-pooled through multilingual-E5-Large (frozen) to yield a 1024-dimensional embedding. Because both encoders are frozen, no gradient flows back into their parameters; embeddings for all 61,773 samples are computed once and cached to disk, decoupling the (expensive) encoder pass from the (cheap, repeatable) classifier-training loop. For fusion, each modality's embedding is independently L2-normalized and the two 1024-dimensional vectors are concatenated into a single 2048-dimensional fused feature vector — an early-fusion design chosen for its established effectiveness in image-text classification, consistent with recent multimodal-fusion surveys (Li and Tang, 2025) and directly validated for this domain by the confirmatory result in Hanif et al. (2026), and for its compatibility with a single downstream classifier head.

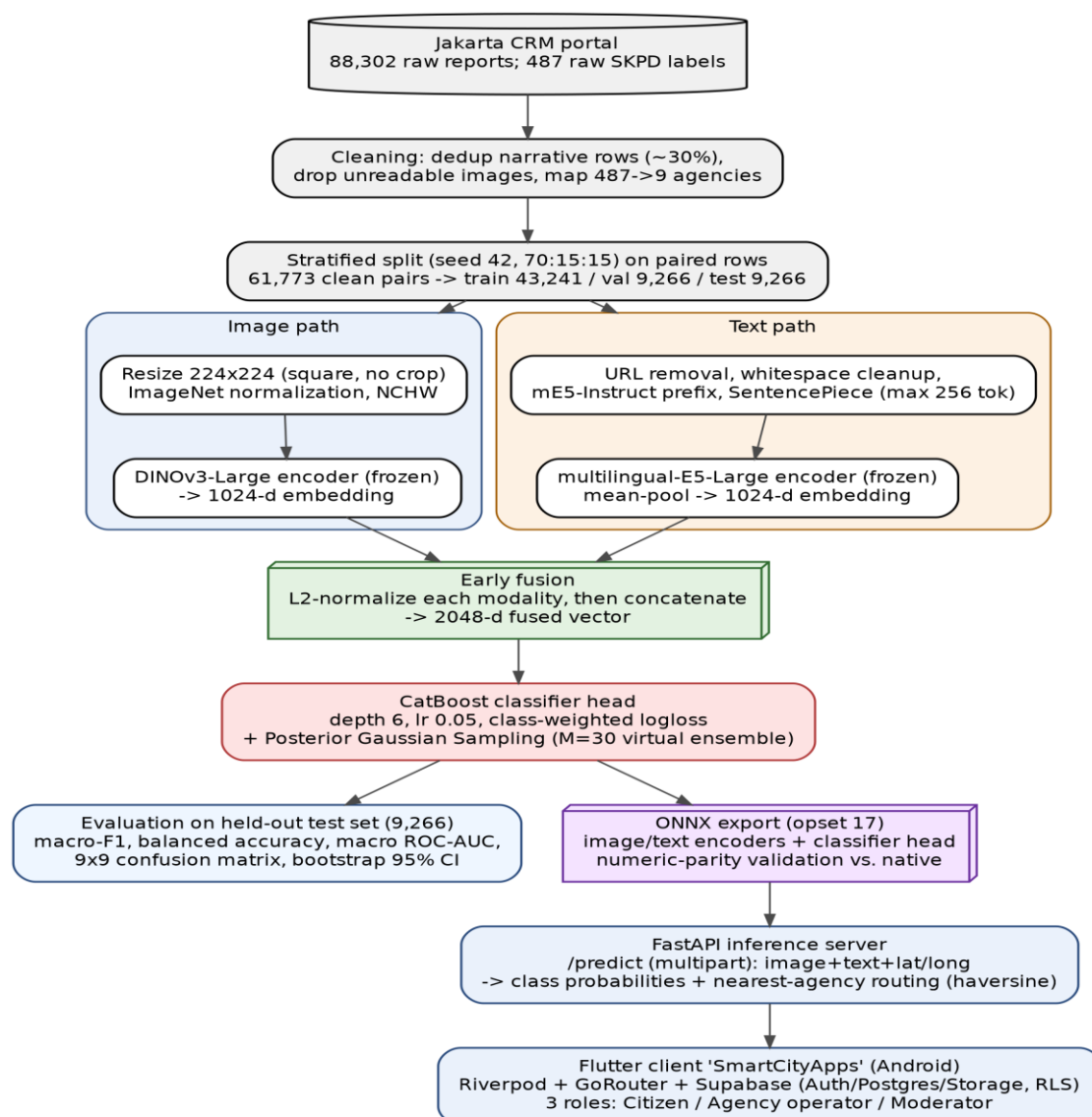


Figure 2. End-to-end multimodal pipeline: data acquisition and cleaning, parallel image/text preprocessing, frozen-encoder feature extraction, early fusion, the CatBoost+PGS classifier head, evaluation, and the ONNX/FastAPI/Flutter deployment path.

2.3 Classifier Head: CatBoost with Posterior Gaussian Sampling

The fused 2048-dimensional vectors are classified using CatBoost (Prokhorenkova et al., 2018), selected as the sole classification algorithm on eight grounds: ordered boosting, which mitigates target leakage and stabilizes generalization on a medium-sized dataset (61,773 samples); oblivious (symmetric) trees, which are fast and CPU-cache-friendly at inference time; documented robustness on high-dimensional tabular inputs such as the 2048-d fusion vector; native posterior sampling via Posterior Gaussian Sampling (Ustimenko et al., 2023), which yields an uncertainty estimate intended for the client interface (its loss at ONNX export is reported in Section 3.5 and its calibration and error-detection value in Section 3.4); deterministic, seed-controlled training that keeps ablation experiments reproducible and auditable; a first-party ONNX export path with high numerical parity to the native implementation; light training cost (minutes per configuration), which made a full 16-way encoder ablation feasible in a single session because both encoders are frozen; and native

handling of categorical features via ordered target statistics, which leaves room for future extensions (e.g., location or time-of-report features) without changing the algorithm.

Key hyperparameters were maximum 1,500 iterations for the non-PGS baseline variant and 3,000 for the PGS variant, tree depth 6, learning rate 0.05, multiclass log-loss objective, L2 leaf regularization 3.0, class weights to offset label imbalance, early stopping on the validation multiclass log-loss (patience of 50 rounds for the baseline and 200 rounds for the PGS variant), and a fixed random seed (42). Under PGS, CatBoost builds a virtual ensemble of $M = 30$ members at inference (Malinin et al., 2021); each member's logits are converted to probabilities with a softmax and the member probabilities are averaged (linear pooling), which is the aggregation behind all PGS results reported here; the study estimates epistemic uncertainty as the mutual information between the prediction and the model's posterior parameters — the difference between the entropy of the mean prediction and the mean of the per-member entropies (Hüllermeier and Waegeman, 2021). The mean per-member entropy is the corresponding aleatoric component, and the two sum to the predictive entropy of the pooled distribution. The mean uncertainty of 0.00339 recorded in the original experiment log is a simpler dispersion score — the across-member standard deviation of the class probabilities, averaged over classes — and is reported as such in Section 3.3.

2.4 Comparative Experimental (Ablation) Design

Two ablation studies were designed to ground the final architecture in evidence rather than assumption. First, a full-matrix ablation trained an identical CatBoost configuration on every combination of four candidate image encoders (DINOv3-Large, Siméoni et al., 2025; DINOv2-Large, Oquab et al., 2024; EVA-02-Large, Fang et al., 2024; Hiera-Large, Ryali et al., 2023) and four candidate text encoders (multilingual-E5-Large-Instruct, Wang et al., 2024; BGE-M3, Chen et al., 2024; IndoBERT-Large-p2, Wilie et al., 2020; Cendol-mT5-Large, Cahyawijaya et al., 2024) — 16 combinations in total — to map each encoder's contribution to final accuracy and select the best pair for deployment. Second, a PGS on/off ablation evaluated the effect of Posterior Gaussian Sampling by comparing accuracy and macro-F1 with and without PGS, together with the mean across-member dispersion of the virtual-ensemble probabilities (Section 2.3).

A methodologically important complication surfaced during this second ablation and is treated in this study as an object of investigation in its own right, not merely a limitation. The baseline and PGS checkpoints were originally trained as two separate runs with different iteration budgets (1,500 vs. $\approx 2,989$ trees), so a naive “with PGS” vs. “without PGS” comparison conflates the PGS mechanism with a near doubling of boosting iterations. To isolate the two effects, a second, confound-controlled evaluation held the trained checkpoint fixed and varied only the inference mode (ordinary argmax vs. PGS-averaged inference with $M = 30$), and a bootstrap resampling procedure (2,000 resamples over the 9,266-sample test set) was used to gauge whether the naively reported gain exceeds sampling variance; because this interval comparison is not a paired test, worst-case McNemar bounds (Section 2.6) and, in the replication run, per-sample paired tests are reported alongside it (Section 3.3). Figure 3 summarizes this design and its outcome.

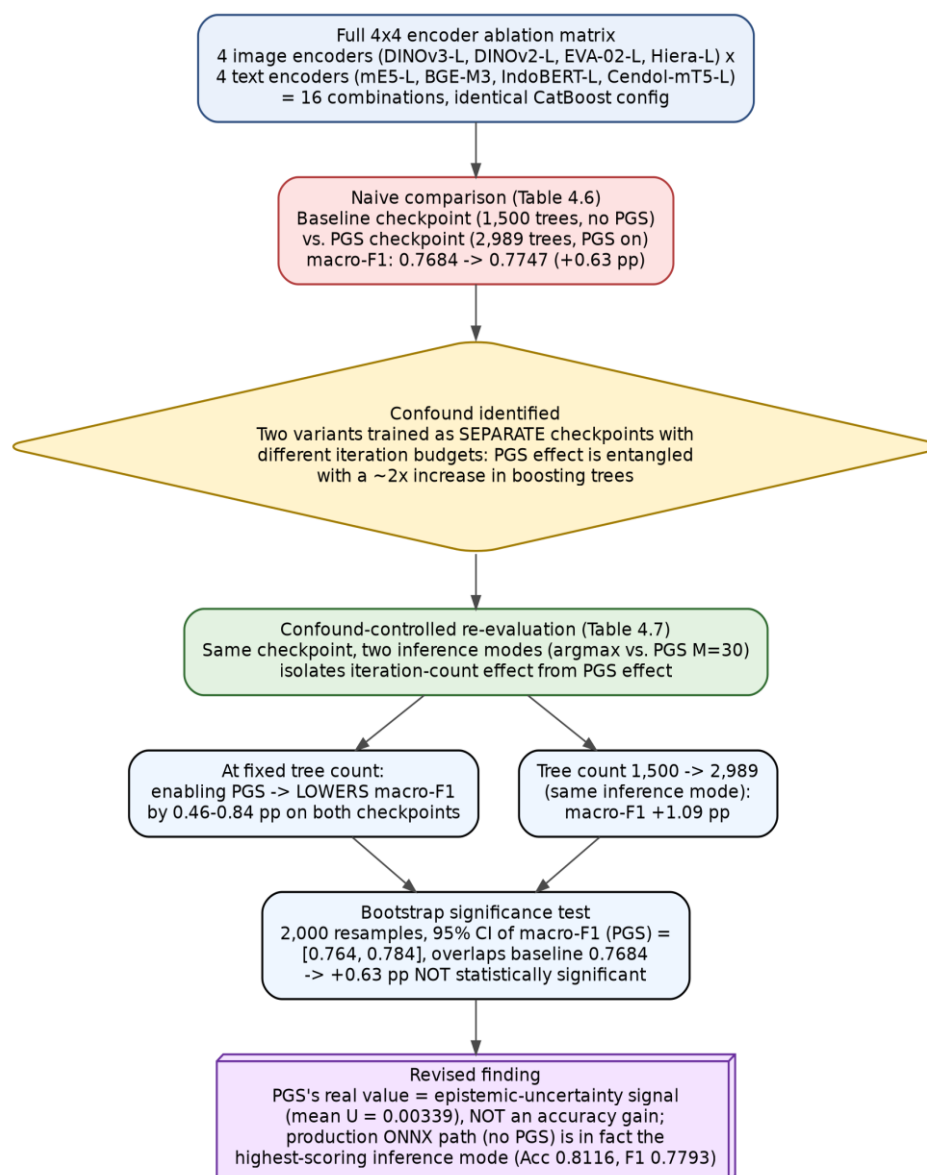


Figure 3. Confound-controlled ablation design isolating the effect of Posterior Gaussian Sampling from the concurrent increase in boosting-tree count, with the bootstrap significance check that overturns the naive comparison.

2.5 ONNX Export and Client-Server Deployment

After the best checkpoint was selected on validation macro-F1, the image encoder, text encoder, and CatBoost classifier head were each exported to ONNX (opset 17). Numerical-parity validation compared native and ONNX outputs on identical inputs, reporting the maximum absolute probability difference ($|\Delta p|_{\max}$) across all output components; the classifier head was validated on the full 9,266-sample test set, while the text encoder's export used Hugging Face Optimum's built-in validation and the image encoder's export parity is reported as not yet re-verified in this analysis environment (see Section 3.5). A structurally important finding from this stage is that CatBoost's PGS virtual-ensemble mechanism has no ONNX operator equivalent and is therefore not exported; the production ONNX graph consequently serves plain (non-PGS) CatBoost probabilities — a discrepancy quantified in Section 3.

The exported artefacts are loaded once at startup by a FastAPI inference server exposing a /predict multipart endpoint that accepts an image file, a report narrative, and latitude/longitude, plus a /health endpoint. On receiving a request, the server reproduces the training-time preprocessing and early fusion path, obtains class probabilities from the classifier head, and performs nearest-agency routing by combining the predicted class with a haversine-distance search over a registry of active agency offices. The Android client, “SmartCityApps,” is built in Flutter with a feature-first architecture, Riverpod state management, GoRouter navigation, and Supabase for authentication, PostgreSQL storage, and file storage, serving three roles (citizen reporter, agency operator, moderator) enforced through Row-Level Security. Separating heavy server-side inference from a lightweight mobile client keeps the Android package free of any on-device model — the ONNX artefacts total roughly 7.4 GB (1.15 GB image encoder, 6.2 GB text encoder, 31 MB classifier head), far beyond a reasonable app-package budget — while allowing centralized model updates without app re-releases.

2.6 Evaluation Metrics

Model performance is reported on the held-out test partition (9,266 samples, never used for training or model selection) using macro-F1 as the primary metric (equal weight per class, appropriate under class imbalance), balanced accuracy (mean per-class recall), macro (one-vs-rest) ROC-AUC, and the normalized 9×9 confusion matrix, together with per-class precision/recall/F1. Statistical robustness of comparative claims is assessed via bootstrap resampling (2,000 iterations) to obtain 95% confidence intervals. Deployment-oriented metrics include per-stage inference latency (CPU, post warm-up, averaged over repeated runs), ONNX-native maximum probability deviation, and end-user usability via a nine-item adaptation of the System Usability Scale (Brooke, 1996), interpreted against standard acceptability, adjective-rating, and percentile benchmarks (Bangor et al., 2008, 2009; Lewis and Sauro, 2018).

Beyond top-1 metrics, the probability outputs and paired differences are analysed on per-sample records. The per-sample predictions of the original run were not archived and its data snapshot can no longer be retrieved; these analyses therefore use a replication run in which the identical pipeline — the label mapping and de-duplication rules of Section 2.1, the stratified 70:15:15 split with seed 42, the frozen encoders and early fusion of Section 2.2, and the CatBoost configuration of Section 2.3 — was re-executed on a later mirror of the same CRM portal for the three strongest image encoders of Figure 4, namely DINOv3-Large (DINOv3-L), EVA-02-Large (EVA-02-L), and DINOv2-Large (DINOv2-L), each paired with multilingual-E5-Large (mE5-L); the non-PGS baseline checkpoint was retrained only for DINOv3-L + mE5-L. From 86,888 raw records carrying 467 raw agency labels, label consolidation and exact de-duplication of narratives leave 37,647 records (exact duplicates make up 49,241 records, 56.7%, of this snapshot against approximately 30% of the original one, which is why the replication data set is smaller despite a similar raw count), and removal of 144 rows whose image could no longer be retrieved yields 37,503 image–text pairs submitted between February 2023 and December 2023 (26,251 training, 5,626 validation, and 5,626 test pairs). DINOv3-L was loaded from the timm port of the same ViT-L/16 LVD-1689M weights, with the DINOv3 processor settings of Section 2.2 re-implemented, because the original weight repository is access-gated. For every validation and test sample, the replication records the nine-class probability vector, the correctness of the top-1 prediction and, for PGS inference, the entropy decomposition of the virtual ensemble. Calibration is

summarised by the top-label expected calibration error (ECE; 15 equal-width confidence bins) with a bootstrap 95% confidence interval, the multiclass Brier score, and the negative log-likelihood (NLL), and is visualised with reliability diagrams (Guo et al., 2017); temperature scaling fitted on the validation partition serves as a post-hoc calibration reference. Two classifiers evaluated on the same test samples are compared with McNemar’s exact test on top-1 correctness (Dietterich, 1998) and with a paired bootstrap of the differences in accuracy and macro-F1 (2,000 resamples of shared sample indices), with Holm’s correction within each family of comparisons (Holm, 1979), applied to the McNemar p-values; paired differences in ECE, Brier score, NLL, and error-detection AUROC are estimated with the same shared bootstrap resamples, and bootstrap intervals are not adjusted for multiplicity. For the original run, where only aggregate accuracies are available, the largest McNemar p-value compatible with them is obtained by evaluating the exact test at the maximum number of discordant pairs allowed by the two error counts (or by the number of top-1 disagreements between the two classifiers, where that number is known); a difference that is significant at this bound is significant for every possible per-sample configuration. Selective prediction is evaluated by ranking the test samples from most to least certain under a given uncertainty score and reporting the accuracy of the accepted reports as a function of coverage, the area under the risk–coverage curve (AURC; Geifman & El-Yaniv, 2017), and the area under the receiver operating characteristic (ROC) curve for detecting misclassified samples (AUROC; Hendrycks & Gimpel, 2017). The scores compared are the maximum class probability and, for PGS inference, the total (predictive entropy), aleatoric (expected member entropy), and epistemic (mutual information) components of the virtual-ensemble uncertainty (Malinin et al., 2021).

3. Results and Discussion

This section reports the results obtained under the protocol described in Section 2 and interprets them against the research objectives and the related literature. Unless stated otherwise, all classification metrics are computed once on the held-out test partition ($n = 9,266$ image–text pairs), which was never used for training, early stopping, or encoder selection; the only exception is Section 3.4, which analyses a replication run on a later data snapshot (test $n = 5,626$; Section 2.6). Table 1 gives an overview of the headline results: for each value it states how the value was obtained and the subsection, table, or figure of this article in which it is presented and discussed.

Table 1. Overview of headline results, how each value was obtained, and where it is reported in this article.

Result	Value	How the value was obtained	Reported in
A. Overall classification performance (method: Sections 2.2, 2.3, 2.6)			
Accuracy	0.8074	Share of correct top-1 predictions of the deployed configuration: DINOv3-L and mE5-L embeddings, L2-normalized and concatenated (2048-d), CatBoost head with PGS ($M = 30$), native inference.	Section 3.1; Table 2
Macro-F1 (primary metric)	0.7747	Unweighted mean of the nine per-class F1 scores, same predictions.	Section 3.1; Table 2
Balanced accuracy	0.7916	Mean of the nine per-class recalls, same predictions.	Section 3.1; Table 2
Macro ROC-AUC (one-vs-rest)	0.9667	Per-class one-vs-rest AUC from the predicted probabilities, averaged over nine classes.	Section 3.1

Result	Value	How the value was obtained	Reported in
95% CI of macro-F1	[0.764, 0.784]	2,000 bootstrap resamples (with replacement) of the test predictions.	Sections 3.1 and 3.3
B. Contribution of each modality and encoder (method: Section 2.4)			
Image-only macro-F1	0.5973	Same CatBoost+PGS head trained on the DINOv3-L embedding only.	Section 3.2; Table 3
Text-only macro-F1	0.7515	Same head trained on the mE5-L embedding only.	Section 3.2; Table 3
Fusion gain	+2.32 pp	Fused macro-F1 minus text-only macro-F1 (0.7747 – 0.7515).	Section 3.2; Table 3
Best encoder pair (of 16)	0.7747	DINOv3-L or EVA-02-L with mE5-L; full 4 × 4 ablation, identical head.	Section 3.2; Figure 4
C. Posterior Gaussian Sampling (method: Section 2.4, Figure 3)			
Naive PGS gain (confounded)	+0.63 pp macro-F1	Two separately trained checkpoints (≈ 2,989 vs. 1,500 trees).	Section 3.3; Table 3
PGS effect at fixed trees	–0.46 to –0.84 pp macro-F1	Same checkpoint evaluated with PGS-averaged vs. ordinary argmax inference.	Section 3.3; Table 4
D. Calibration, paired tests, and selective prediction (method: Section 2.6)			
Fusion gain, paired test	$p \leq 0.003$	Exact McNemar test at the least favourable split of discordant pairs compatible with the accuracies in Table 3 (CatBoost+PGS head; $p \leq 0.031$ without PGS).	Section 3.2
PGS averaging at fixed trees, paired test	$p \leq 7.6 \times 10^{-4}$	Same bound, using the 130 top-1 disagreements between the two inference modes of the PGS checkpoint.	Section 3.3
Top-label ECE, served path vs. PGS-averaged	0.0271 vs. 0.0379	15-bin expected calibration error of the PGS checkpoint under ordinary argmax vs. PGS-averaged inference, from per-sample records of the replication run (test $n = 5,626$).	Section 3.4; Table 5; Figure 5
DINOv3-L vs. EVA-02-L, paired test	Δ macro-F1 = –0.60 pp [–1.33, 0.18]; McNemar (top-1 correctness) Holm-adjusted $p = 0.257$	Paired bootstrap of Δ macro-F1 and exact McNemar test on identical test samples of the replication run (mE5-L, CatBoost+PGS).	Section 3.4; Table 6
Error-detection AUROC, maximum probability vs. mutual information	0.8041 vs. 0.6447	Area under the ROC curve for separating incorrect from correct test predictions in the replication run (served path vs. PGS epistemic score).	Section 3.4; Table 7; Figure 6
E. ONNX export (method: Section 2.5)			
Classifier-head parity, $ \Delta p _{\max}$	1.38×10^{-6}	Native CatBoost (without PGS) vs. ONNX (opset 17), all 9 classes × 9,266 samples.	Section 3.5; Table 8
Top-1 agreement	9,266 / 9,266	Samples with identical top-1 class under native and ONNX inference.	Section 3.5; Table 8
F. Deployed system (method: Section 2.5)			
Nearest-agency routing	45 / 45 routes	9 predicted classes × 5 landmark test points (one per administrative city); median 6.90 km, max 11.83 km.	Section 3.6; Table 9
End-to-end latency (CPU)	1.08 s	Sum of per-stage ONNX latencies for one sample after warm-up.	Section 3.6; Table 10
Client package size	216 MB vs. 7.4 GB	Android debug APK without any model vs. total server-side ONNX artefacts.	Section 3.6
G. Users (method: Section 2.6)			
Usability (SUS)	71.6 (acceptable)	Nine-item SUS adaptation, $n = 30$ citizen reporters; Cronbach's $\alpha = 0.74$.	Section 3.7; Table 11; Figure 7

Notes: Groups A–C and E and the first two rows of group D are computed on the held-out test partition of the original run ($n = 9,266$); the last three rows of group D come from the replication run described in Section 2.6 (test $n = 5,626$). Group headings give the Methods subsection describing each procedure; the last column gives the Results subsection, table, or figure where the value is presented and discussed. PGS = Posterior Gaussian Sampling; DINOv3-L = DINOv3-Large; mE5-L = multilingual-E5-Large; CI = confidence interval; pp = percentage points; SUS = System Usability Scale; EVA-02-L = EVA-02-Large; ECE = expected calibration error; AUROC = area under the receiver operating characteristic curve.

3.1 Overall Classification Performance

The deployed configuration (frozen DINOv3-Large and multilingual-E5-Large embeddings, early fusion, and a CatBoost head with PGS) reaches an accuracy of 0.8074 and a macro-F1 of 0.7747 on the test set, with a 95% bootstrap confidence interval for macro-F1 of [0.764, 0.784], a balanced accuracy of 0.7916, a macro one-vs-rest ROC-AUC of 0.9667, and a weighted F1 of 0.8061. These four views of the same predictions should be read together on a nine-class, imbalanced task. The gap of roughly three percentage points between accuracy and macro-F1 directly quantifies how much the dominant classes inflate the aggregate score; the balanced accuracy of 0.7916 shows that the model does not collapse onto majority classes; and the high ROC-AUC indicates that, independently of the argmax threshold, the correct agency is usually ranked above the incorrect ones, so most residual errors concern samples lying close to a decision boundary rather than a wholesale failure to separate classes. Training one CatBoost configuration took on average 308 s without PGS and 602 s with PGS, which is what made the exhaustive encoder ablation in Section 3.2 affordable.

Table 2 breaks the result down by agency. Classes with large support and distinctive vocabulary or visual content perform best, notably Dinas Pertamanan dan Hutan (parks and urban forestry; $F1 = 0.9022$) and Dinas Bina Marga (roads; $F1 = 0.8720$). The residual “Other agencies” class is the weakest ($F1 = 0.5152$) because it aggregates heterogeneous reports without a consistent visual or textual signature, followed by Dinas Sumber Daya Air (water resources; $F1 = 0.6367$), whose low precision (0.5577) reflects flooding and drainage reports that legitimately overlap with road and neighbourhood-level complaints. Class scarcity alone does not explain performance: Dinas Cipta Karya, Tata Ruang, dan Pertanahan reaches $F1 = 0.8584$ with only 221 test samples, and Badan Pembinaan BUMD reaches 0.7273 with 252. Errors are therefore driven more by semantic overlap between agency mandates than by sample count, a point analysed further in Section 3.8.

Table 2. Per-class precision, recall, and F1 of the deployed configuration on the test set (DINOv3-L + mE5-L + CatBoost+PGS).

Agency (class)	Precision	Recall	F1	Support
Dinas Bina Marga (roads)	0.8644	0.8797	0.8720	2,029
Satuan Polisi Pamong Praja (public order)	0.8432	0.8699	0.8564	1,261
Dinas Perhubungan (transportation)	0.8443	0.8699	0.8569	1,752
Kelurahan (urban village office)	0.8103	0.6939	0.7476	1,797
Dinas Pertamanan dan Hutan (parks and forestry)	0.8924	0.9122	0.9022	900
Dinas Sumber Daya Air (water resources)	0.5577	0.7417	0.6367	391
Dinas Cipta Karya, Tata Ruang, dan Pertanahan	0.8398	0.8778	0.8584	221
Badan Pembinaan BUMD	0.6711	0.7937	0.7273	252
Other agencies (residual class)	0.5486	0.4857	0.5152	663
Macro average	0.7635	0.7916	0.7747	9,266
Weighted average	0.8086	0.8073	0.8061	9,266

Note: the macro-average recall equals the balanced accuracy.

3.2 Contribution of Each Modality and Encoder Selection

Table 3 isolates the contribution of each modality by training the identical CatBoost head on the image embedding only, the text embedding only, and the fused vector. Text is by far the more discriminative modality (macro-F1 0.7515 versus 0.5973 with PGS), which is expected because citizen narratives often name the problem type or even the responsible agency, whereas a photograph of, for example, standing water is visually ambiguous between drainage, road, and neighbourhood-level mandates. Early fusion nevertheless improves on the strongest single modality in both head variants: by 1.86 percentage points of macro-F1 without PGS and by 2.32 points with PGS, and by 17.74 points over the image-only model. Because the text-only macro-F1 (0.7515) lies below the lower bound of the fused model's 95% bootstrap interval (0.764), the fusion gain is unlikely to be an artefact of sampling variance, and a paired test confirms it: McNemar's exact test, evaluated at the least favourable split of discordant pairs compatible with the accuracies in Table 3 (Section 2.6), still gives $p \leq 0.003$ for the head with PGS and $p \leq 0.031$ for the head without PGS. The image therefore carries complementary evidence that the narrative does not always express, which is the practical justification for asking citizens to attach a photograph.

Table 3. Modality ablation on the test set (image encoder DINOv3-L, text encoder mE5-L).

Configuration	Accuracy (CatBoost)	Macro-F1 (CatBoost)	Accuracy (CatBoost+PGS)	Macro-F1 (CatBoost+PGS)
Image only (DINOv3-L)	0.6265	0.6016	0.6238	0.5973
Text only (mE5-L)	0.7850	0.7498	0.7874	0.7515
Early fusion (image + text)	0.7996	0.7684	0.8074	0.7747
Fusion gain over text only	+1.46 pp	+1.86 pp	+2.00 pp	+2.32 pp

pp = percentage points. The "with" and "without PGS" columns come from two separately trained checkpoints and are not a controlled comparison of PGS (see Section 3.3).

The 4×4 encoder ablation in Figure 4 shows two regularities. First, multilingual-E5-Large is the best text encoder in every row, ahead of BGE-M3 (Chen *et al.*, 2024), IndoBERT-Large-p2 (Wilie *et al.*, 2020), and Cendol-mT5-Large (Cahyawijaya *et al.*, 2024). This agrees with recent evidence that E5-family embeddings generalise well across complaint categories in public-service text (Rakhimzhanov *et al.*, 2025), and suggests that contrastive sentence-embedding pre-training matters more here than language-specific pre-training, since the Indonesian-only IndoBERT and the generative Cendol model both trail the multilingual embedding models. Second, the choice of text encoder matters roughly four times more than the choice of image encoder: with DINOv3-L fixed, macro-F1 spans 0.7340–0.7747 across text encoders (4.07 pp), whereas with mE5-L fixed it spans only 0.7652–0.7747 across image encoders (0.95 pp). Among image encoders, DINOv3-L and EVA-02-L (Fang *et al.*, 2024) tie at 0.7747, DINOv2-L (Oquab *et al.*, 2024) is marginally lower at 0.7736, and the efficiency-oriented Hiera-L (Ryali *et al.*, 2023) is the weakest in all four pairings. Because the top two image encoders are indistinguishable on accuracy, the selection of DINOv3-L for deployment rests on recency and licensing/maintenance considerations rather than on a measured accuracy advantage, and we report it as such. In the replication run of Section 3.4, where EVA-02-L is numerically ahead by 0.60 pp macro-F1, a paired test on identical test samples finds no significant difference (Table 6); this does not establish that the two encoders are equivalent.

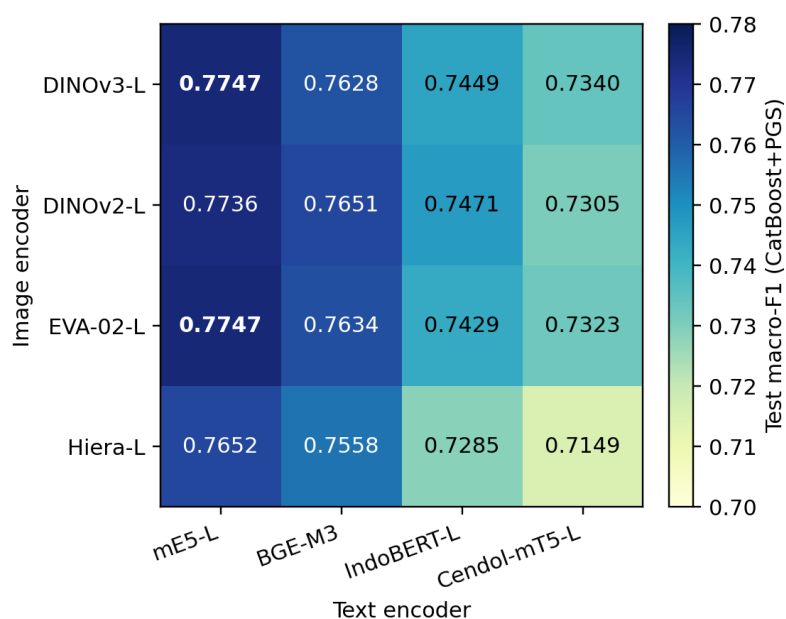


Figure 4. Test macro-F1 of CatBoost+PGS for all 16 image-encoder $\times$ text-encoder pairings. Bold cells mark the best result (0.7747).

3.3 Confound-Controlled Re-evaluation of Posterior Gaussian Sampling

A naive comparison of the two trained checkpoints suggests that PGS improves accuracy by 0.78 pp and macro-F1 by 0.63 pp (Table 3, fusion row). As described in Section 2.4, however, the PGS checkpoint stopped at 2,989 trees and the baseline at 1,500, so the comparison conflates the PGS mechanism with a near-doubling of boosting iterations. Table 4 separates the two factors by evaluating each checkpoint under both inference modes.

Table 4. Separation of the tree-count effect from the PGS effect (DINOv3-L + mE5-L, test set, $n = 9,266$).

Checkpoint and inference mode	Trees	Accuracy	Macro-F1
Baseline checkpoint, ordinary argmax	1,500	0.7996	0.7684
Baseline checkpoint, PGS-averaged inference ($M = 30$)	1,500	0.7914	0.7600
PGS checkpoint, ordinary argmax	2,989	0.8116	0.7793
PGS checkpoint, PGS-averaged inference ($M = 30$)	2,989	0.8073	0.7747
Effect of PGS at fixed trees (row 2 – row 1; row 4 – row 3)	—	–0.82; –0.43 pp	–0.84; –0.46 pp
Effect of trees at fixed inference mode (row 3 – row 1)	—	+1.20 pp	+1.09 pp

pp = percentage points; M = number of virtual-ensemble members.

Once the tree count is held fixed, enabling PGS inference lowers performance on both checkpoints, by 0.43–0.82 pp in accuracy and 0.46–0.84 pp in macro-F1, while the whole apparent improvement is explained by the additional boosting iterations (+1.20 pp accuracy and +1.09 pp macro-F1 under the same inference mode). For the PGS checkpoint this loss is statistically significant even without per-sample records: the two inference modes disagree on the top-1 class for only 130 of 9,266 test samples (Section 3.5), and McNemar’s exact test evaluated at the least favourable split of these disagreements that is compatible with the reported accuracies still gives $p \leq 7.6 \times 10^{-4}$. The bootstrap analysis points the same way: the baseline macro-F1 (0.7684) falls inside the 95% interval of the PGS

configuration ([0.764, 0.784]), which suggests, but cannot establish, that even the naive +0.63 pp difference lies within sampling variance. Because both configurations are evaluated on the same test samples, this interval-overlap argument is not a valid paired test: from the aggregate accuracies, the exact McNemar p-value of the naive comparison can only be bounded from above ($p \leq 0.25$), so the original run neither establishes nor excludes a real difference, which would in any case reflect how the two checkpoints were trained (a larger iteration budget and, for the PGS checkpoint, posterior-sampling training with Langevin noise and model shrinkage) rather than PGS-averaged inference. In the paired test of the replication run, the naive difference is not significant (Section 3.4, Table 6). The claim that PGS improves classification accuracy is therefore not supported by these data. This is an instance of a more general reproducibility problem in machine-learning benchmarking, in which differences in training budget, early-stopping outcome, or random seed are attributed to the method under study (Bouthillier *et al.*, 2021).

The small and, under control, negative effect is consistent with the nature of the method and of the data. PGS forms a virtual ensemble from a single CatBoost model (Ustimenko *et al.*, 2023), so its members share parameters and are strongly correlated, unlike independently trained deep ensembles (Lakshminarayanan *et al.*, 2017); its posterior mean therefore differs from the point prediction only near decision boundaries, which the high ROC-AUC shows to be a small fraction of samples. More fundamentally, the dominant uncertainty in this task is aleatoric rather than epistemic: it originates in the many-to-one mapping of 487 raw agency names onto nine classes, in very short narratives, and in reports that could legitimately be routed to more than one agency. Aleatoric uncertainty is irreducible and cannot be removed by averaging ensemble members (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021), which is consistent with the very small disagreement between the virtual-ensemble members (mean across-member standard deviation of the class probabilities, 0.00339). The same pattern appears in the encoder matrix, where PGS-averaged macro-F1 is nearly flat across competent image encoders (0.7747, 0.7747, and 0.7736). The documented value of PGS in this system therefore lies, if anywhere, in an uncertainty signal at negligible inference cost (about 0.05 ms per sample for the head with 30 virtual members) rather than in an accuracy gain, and the replication in Section 3.4 shows that even this signal is weaker than the maximum class probability of the argmax (served) inference path; if stronger reliability guarantees are required, coverage-guaranteed selective prediction or set-valued outputs through conformal prediction (Angelopoulos & Bates, 2023) are more appropriate than enlarging the virtual ensemble (see also Abdar *et al.*, 2021; Gawlikowski *et al.*, 2023).

3.4 Calibration, Paired Significance Tests, and Selective Prediction

Sections 3.1–3.3 evaluate the top-1 decisions of the original run, whereas the value attributed to PGS in Section 3.3 lies in its uncertainty signal. This subsection evaluates the probability outputs themselves on the replication run described in Section 2.6 (test $n = 5,626$). Because the snapshot, sample size, and class distribution differ from the original data, the replication is an independent re-run of the same design rather than a reproduction of Tables 2–4, and it doubles as a check of the original conclusions on a later snapshot of the same portal. Retrained on the replication data and evaluated on its test partition, the deployed configuration reaches an accuracy of 0.8066 and a macro-F1 of 0.7721 with PGS-averaged inference (original run: 0.8074 and 0.7747) and 0.8102 and 0.7752 with ordinary argmax on the same checkpoint (original: 0.8116 and 0.7793); with the same head, EVA-02-L and DINOv2-L reach macro-F1 values of 0.7781 and 0.7730 (original: 0.7747 and 0.7736). Three

questions are addressed: whether the predicted confidences are calibrated, whether the encoder ranking in Figure 4 and the PGS comparisons in Section 3.3 hold under paired tests on identical samples, and whether the uncertainty signal identifies the reports that should be deferred to a human operator.

Table 5 reports the top-label calibration of the DINOv3-L + mE5-L head retrained in the replication run, for both checkpoints and both inference modes, and Figure 5 shows reliability diagrams for its PGS checkpoint. Ordinary argmax inference on the PGS checkpoint is the inference mode of the ONNX head served in production (Section 3.5) and is referred to below as the served (argmax) path. At the fixed PGS checkpoint, PGS-averaged inference ($M = 30$) worsens calibration relative to the served argmax path: the ECE is 0.0379 versus 0.0271, a paired difference of +0.0107 (95% CI [0.0054, 0.0163]). The two strictly proper scores rise as well: the Brier score from 0.2822 to 0.2900 (paired difference +0.0079, 95% CI [0.0071, 0.0087]); the NLL from 0.5779 to 0.5965 nats (paired difference +0.0185, 95% CI [0.0169, 0.0200]). Both inference paths are underconfident on average, with mean confidence falling short of accuracy by 2.66 pp (served argmax) and 3.77 pp (PGS-averaged). Temperature scaling fitted on the validation partition ($T = 0.90$) leaves every top-1 prediction unchanged and lowers the ECE of the served path to 0.0204, but not significantly (paired difference -0.0068 , 95% CI $[-0.0152, 0.0039]$); it does lower the Brier score (paired difference -0.0015 , 95% CI $[-0.0023, -0.0007]$) and the NLL (paired difference -0.0046 , 95% CI $[-0.0068, -0.0021]$). Paired differences are computed from unrounded values and may differ in the last digit from differences of the rounded entries in Table 5.

Table 5. Calibration of the DINOv3-L + mE5-L classifier head on the replication test set ($n = 5,626$) by checkpoint and inference mode.

Checkpoint and inference mode	Accuracy	Mean confidence	ECE [95% CI]	Brier score	NLL
Baseline checkpoint, ordinary argmax	0.8027	0.7614	0.0441 [0.0354, 0.0540]	0.2939	0.6076
Baseline checkpoint, PGS-averaged ($M = 30$)	0.7972	0.7467	0.0517 [0.0430, 0.0614]	0.3026	0.6284
PGS checkpoint, ordinary argmax (served path)	0.8102	0.7835	0.0271 [0.0214, 0.0381]	0.2822	0.5779
PGS checkpoint, PGS-averaged ($M = 30$; inference mode of the headline results)	0.8066	0.7689	0.0379 [0.0319, 0.0493]	0.2900	0.5965
PGS checkpoint, ordinary argmax + temperature scaling	0.8102	0.8100	0.0204 [0.0164, 0.0314]	0.2807	0.5734

ECE = top-label expected calibration error (15 equal-width confidence bins; 95% CI from 2,000 bootstrap resamples); Brier score = multiclass Brier score (sum over the nine classes; 0 = perfect); NLL = negative log-likelihood in nats. Temperature scaling uses a single temperature fitted on the validation partition ($T = 0.90$). In the replication run the baseline checkpoint has 1,494 trees and the PGS checkpoint 2,289 trees (original run: 1,500 and 2,989; Table 4).

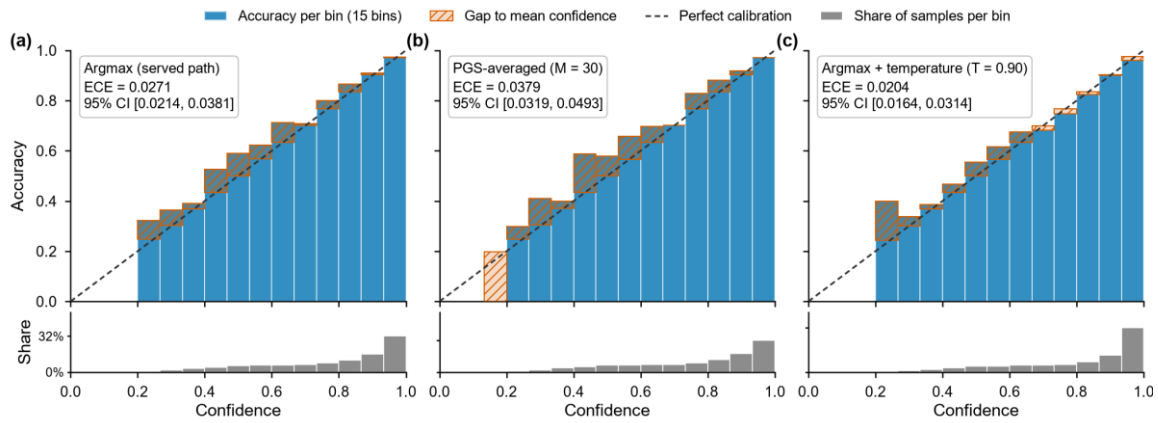


Figure 5. Reliability diagrams of the DINOv3-L + mE5-L head on the replication test set: (a) PGS checkpoint with ordinary argmax inference (the served path), (b) the same checkpoint with PGS-averaged inference ($M = 30$), and (c) ordinary argmax after temperature scaling fitted on the validation partition. Bars show the accuracy in each of 15 confidence bins; the diagonal marks perfect calibration.

Table 6 complements the encoder tie of Section 3.2 and the interval-overlap argument of Section 3.3, both from the original run, with paired tests on identical test samples of the replication run. Among the image encoders, each fused with mE5-L and evaluated with PGS-averaged inference, none of the three pairwise differences in top-1 correctness is significant after Holm adjustment (McNemar's exact test; DINOv3-L vs. EVA-02-L: $\Delta\text{accuracy} = -0.50$ pp, adjusted $p = 0.257$; DINOv3-L vs. DINOv2-L: $\Delta\text{accuracy} = +0.20$ pp, adjusted $p = 0.568$; EVA-02-L vs. DINOv2-L: $\Delta\text{accuracy} = +0.69$ pp, adjusted $p = 0.0834$, although the unadjusted $p = 0.0278$ and the unadjusted 95% interval $[0.05, 1.28]$ exclude a zero difference), and all three paired-bootstrap intervals for $\Delta\text{macro-F1}$ include zero (-0.60 , -0.08 , and $+0.52$ pp; Table 6). EVA-02-L is thus numerically ahead of DINOv3-L in this run, and a non-significant difference is not evidence of equivalence. At the fixed PGS checkpoint, PGS-averaged inference changes accuracy by -0.36 pp (95% CI $[-0.62, -0.09]$) and macro-F1 by -0.31 pp (95% CI $[-0.69, 0.06]$) relative to the served argmax path; with 19 reports correct only under PGS averaging and 39 only under argmax, the accuracy difference is statistically significant (McNemar $p = 0.0119$; Holm-adjusted $p = 0.0239$), whereas the macro-F1 interval includes zero. For the naive comparison between the PGS configuration and the baseline checkpoint with argmax inference, the replication gives a smaller difference than the original run ($+0.63$ pp macro-F1 there; here $+0.27$ pp macro-F1, 95% CI $[-0.35, 0.93]$, and $+0.39$ pp accuracy), which is not statistically significant in top-1 correctness (McNemar $p = 0.168$, Holm-adjusted $p = 0.168$); the replication checkpoints also differ less in tree count than the original ones (Table 5 note).

Table 6. Paired comparisons on the replication test set ($n = 5,626$): top image encoders paired with mE5-L under CatBoost+PGS (first three rows) and inference-mode comparisons for DINOv3-L + mE5-L (last two rows).

Comparison (A vs. B)	$\Delta\text{Accuracy}$, pp [95% CI]	$\Delta\text{Macro-F1}$, pp [95% CI]	b / c	McNemar p	Holm-adjusted p
DINOv3-L vs. EVA-02-L	-0.50 $[-1.07, 0.14]$	-0.60 $[-1.33, 0.18]$	144 / 172	0.129	0.257
DINOv3-L vs. DINOv2-L	$+0.20$ $[-0.44, 0.84]$	-0.08 $[-0.81, 0.64]$	159 / 148	0.568	0.568
EVA-02-L vs. DINOv2-L	$+0.69$ $[0.05, 1.28]$	$+0.52$ $[-0.28, 1.27]$	169 / 130	0.0278	0.0834
PGS checkpoint: PGS-averaged vs. ordinary argmax (fixed trees)	-0.36 $[-0.62, -0.09]$	-0.31 $[-0.69, 0.06]$	19 / 39	0.0119	0.0239

Comparison (A vs. B)	Δ Accuracy, pp [95% CI]	Δ Macro-F1, pp [95% CI]	b / c	McNemar p	Holm-adjusted p
PGS checkpoint, PGS-averaged vs. baseline checkpoint, ordinary argmax (naive comparison)	+0.39 [-0.11, 0.92]	+0.27 [-0.35, 0.93]	127 / 105	0.168	0.168

$\Delta = A - B$ in percentage points with 95% paired-bootstrap CI (2,000 resamples); b = samples correct under A only; c = samples correct under B only; McNemar p = exact two-sided binomial test on b and c. Holm adjustment within each family (three encoder comparisons; two inference-mode comparisons).

Table 7 and Figure 6 evaluate how well each uncertainty score separates correct from incorrect predictions and what accuracy the automatically routed reports reach when the least certain ones are deferred to an operator. The maximum class probability of the served argmax path is the best error detector (AUROC 0.8041 [0.7902, 0.8175]), slightly but significantly above the same score computed from the PGS-averaged probabilities (AUROC 0.7985 [0.7849, 0.8122]; paired difference -0.0056 , 95% CI $[-0.0105, -0.0009]$), whereas the epistemic mutual information is a much weaker detector (AUROC 0.6447 [0.6273, 0.6616]; paired difference to the served score -0.1594 , 95% CI $[-0.1749, -0.1444]$); the aleatoric and total entropy terms lie in between (AUROC 0.7811 and 0.7810). Deferring the 20% least certain reports according to the served maximum probability raises the accuracy of the accepted 80% to 0.8809 (from 0.8102 at full coverage), and automatic routing can cover up to 72.4% of the reports while the accepted ones keep at least 90% accuracy.

Table 7. Error detection and selective prediction on the replication test set for alternative uncertainty scores of the DINOv3-L + mE5-L head (PGS checkpoint).

Uncertainty score (inference mode)	AUROC for error detection [95% CI]	AURC ($\times 100$)	Accuracy at 80% coverage	Coverage at 90% accuracy
1 – maximum probability (ordinary argmax; served path)	0.8041 [0.7902, 0.8175]	6.78	0.8809	0.724
1 – maximum probability (PGS-averaged)	0.7985 [0.7849, 0.8122]	7.12	0.8745	0.710
Predictive entropy, total (PGS-averaged)	0.7810 [0.7672, 0.7952]	7.52	0.8649	0.679
Expected member entropy, aleatoric (PGS-averaged)	0.7811 [0.7673, 0.7954]	7.52	0.8647	0.679
Mutual information, epistemic (PGS-averaged)	0.6447 [0.6273, 0.6616]	11.70	0.8258	0.424
Across-member standard deviation of the class probabilities, score of Sections 2.3 and 3.3 (PGS-averaged)	0.6959 [0.6800, 0.7114]	9.89	0.8345	0.527

AUROC = area under the ROC curve for detecting misclassified samples (0.5 = chance); AURC = area under the risk–coverage curve (lower is better); accuracy at 80% coverage = accuracy of the 80% most certain reports; coverage at 90% accuracy = largest share of reports that can be routed automatically while the accepted reports keep at least 90% accuracy. Correctness refers to the predictions of the stated inference mode; 95% CIs from 2,000 bootstrap resamples.

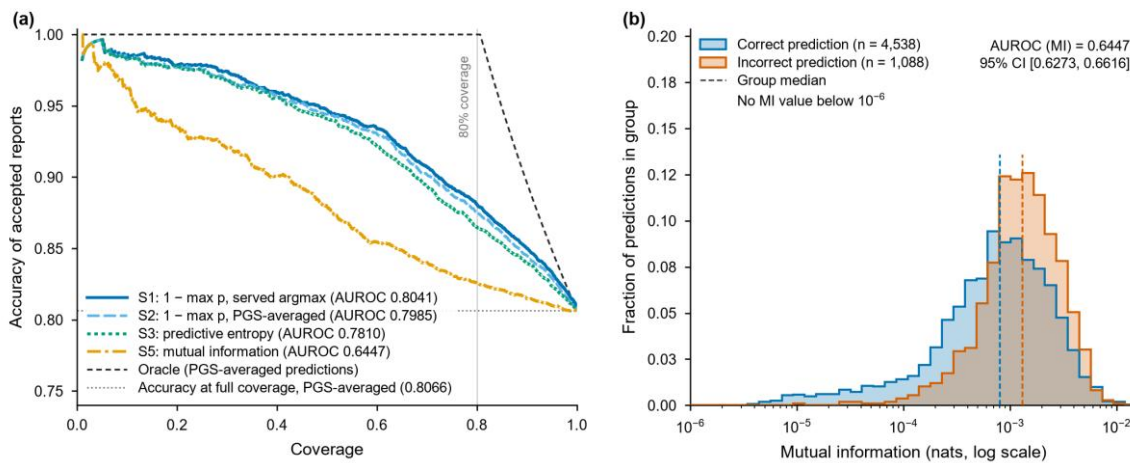


Figure 6. Selective prediction on the replication test set: (a) accuracy of the automatically routed reports as a function of coverage when the least certain reports are deferred, for four uncertainty scores labelled S1, S2, S3, and S5 after rows 1, 2, 3, and 5 of Table 7 (dashed: oracle ranking); (b) distribution of the epistemic mutual information (MI) for correct and incorrect PGS-averaged predictions (logarithmic axis).

Taken together, the replication qualifies the suggestion of Section 3.3 that the value of PGS, if any, lies in its uncertainty signal rather than in accuracy: at a fixed checkpoint, PGS averaging lowers accuracy (-0.36 pp; McNemar $p = 0.0119$) and worsens calibration, and although its mutual information carries an error signal (AUROC 0.6447), the maximum class probability of the served argmax path — which needs no PGS and survives the ONNX export — flags misrouted reports far better (AUROC 0.8041). Operationally, a deferral rule on this score is the practical uncertainty mechanism for the prototype; because the deferral gains in Table 7 come from the replication head, the threshold for the deployed head should be set on its own validation data.

3.5 ONNX Export Parity and the Train–Serving Gap

Table 8 summarises the export validation. For the classifier head, the maximum absolute probability difference between native CatBoost (without PGS) and ONNX Runtime over all nine classes and all 9,266 test samples is 1.38×10^{-6} , one order of magnitude below the pre-set tolerance of 10^{-5} , and the top-1 prediction is identical for every sample. The export of the head therefore introduces no divergence that could change a routing decision. The evidence for the two encoders is weaker and is stated as such: the text-encoder export passed Hugging Face Optimum’s built-in validation, which provides a pass/fail guarantee but no reportable $|\Delta p|_{\max}$, and the image-encoder export has not yet been re-verified numerically because the roughly 7.4 GB of ONNX artefacts were not available in the analysis environment.

Table 8. Export parity and inference-path comparison on the deployed checkpoint (test set, $n = 9,266$).

Inference path	Accuracy	Macro-F1	Parity evidence
Native CatBoost + PGS ($M = 30$) — reported headline	0.8073	0.7747	Reference for headline metrics
Native CatBoost without PGS	0.8116	0.7793	Reference for ONNX parity
ONNX classifier head (served in production)	0.8116	0.7793	$ \Delta p _{\max} = 1.38 \times 10^{-6}$; top-1 identical 9,266/9,266
Text encoder (mE5-L) export	—	—	Passed Optimum validation (pass/fail only)

Inference path	Accuracy	Macro-F1	Parity evidence
Image encoder (DINOv3-L) export	—	—	Not yet re-verified

Top-1 predictions of the native PGS path and the ONNX path differ on 130 of 9,266 samples.

The more consequential finding is that CatBoost’s PGS virtual ensemble has no ONNX operator equivalent and is silently dropped at export. The system that is actually served therefore produces non-PGS probabilities: on 130 of 9,266 test samples its top-1 prediction differs from the PGS path that is reported as the headline result, and its accuracy and macro-F1 (0.8116 and 0.7793) are 0.43 and 0.46 pp higher. The direction of the discrepancy is favourable, which makes the headline figures conservative, but it also means that the epistemic-uncertainty signal that motivated PGS is not available in the running prototype. The replication in Section 3.4 suggests that this loss is less consequential than it appears: under ordinary argmax inference, the mode that is served, the maximum class probability is better calibrated than the PGS-averaged probabilities and a markedly stronger error detector than the PGS epistemic signal. This is a concrete case of train–serving skew, a recurrent failure mode in deployed machine-learning systems (Paley et al., 2022). Two practical remedies follow: either report the metrics of the path that is actually served, or compute the uncertainty in a native CatBoost side-car next to the ONNX head. In either case, parity should be validated for the full inference contract, including post-processing and uncertainty, and not only for the network weights.

3.6 System-Level Validation: Routing, Latency, and Device Footprint

The routing module was verified by running the production routing code on every combination of the nine predicted classes and five public landmarks representing Jakarta’s five administrative cities (45 routes); landmarks were used because the dataset partitions store no coordinates. All 45 routes returned a destination office, with a median haversine distance of 6.90 km and a maximum of 11.83 km. Routing is genuinely coordinate-driven rather than a fixed class-to-office lookup: for Dinas Bina Marga, a report from Monas (Central Jakarta) is routed to the provincial office 1.92 km away, whereas a report from Blok M (South Jakarta) is routed to the South Jakarta sub-office 1.15 km away. Table 9 shows, however, that the sharpness of routing is bounded by the completeness of the office registry, not by the algorithm. Three of the nine classes have only one registered office, so distance plays no role for them, and no office is registered in North Jakarta, so reports from the Tanjung Priok test point are routed 8.97–11.83 km to Central Jakarta. Because the registry is maintained by moderators through the application, extending coverage requires neither code changes nor retraining; replacing straight-line distance with jurisdiction-based routing on administrative boundaries would align assignments more closely with actual agency mandates.

Table 9. Office-registry coverage per class and its effect on nearest-office routing (16 active offices).

Class	Registered offices	Routing behaviour
Dinas Bina Marga	3	Discriminative (distance-dependent)
Satpol PP; Dinas Perhubungan; Kelurahan; Dinas Pertamanan dan Hutan; Dinas Sumber Daya Air	2 each	Discriminative (distance-dependent)
Dinas Cipta Karya, Tata Ruang, dan Pertanahan; Badan Pembinaan BUMD	1 each	Single destination; distance has no effect
Other agencies	1	Manual-review unit (fallback)

Routing verified on 9 classes × 5 landmark test points (45 routes).

Table 10 reports per-stage latency of the ONNX pipeline for one sample on a development CPU after warm-up. End-to-end inference takes about 1.08 s, of which more than 99% is spent in the two encoders (437 ms for the image and 639 ms for the text encoder), whereas the classifier head needs only 0.3 ms. Latency is therefore governed by embedding extraction, not by the classifier, and throughput scales with encoder capacity. Serving the models on the server keeps the client light: the Android package contains no model (a 216 MB debug build), compared with about 7.4 GB of ONNX artefacts (1.15 GB image encoder, 6.2 GB text encoder, and 31 MB classifier head) that on-device inference would require. The trade-off is a dependency on network access and on a single inference endpoint; availability and operating cost were not measured and are treated as limitations (see Acknowledged Limitations).

Table 10. Per-stage inference latency of the served ONNX pipeline (one sample, CPU, after warm-up).

Pipeline stage	Latency (ms)	Share of total
Image encoder (DINOv3-L, ONNX)	437	40.6%
Text encoder (mE5-L, ONNX)	639	59.4%
Classifier head (CatBoost, ONNX)	0.3	< 0.1%
End-to-end	1,076	100%

Latency averaged over 20 measurement iterations after warm-up on a development CPU.

3.7 Usability Evaluation

Usability of the citizen-reporter role was evaluated with 30 respondents who first attempted eight task scenarios covering the core reporting flow (sign-in, photo capture or selection, narrative entry, location setting, AI classification review, manual category correction, and submission) and then completed a questionnaire based on the Indonesian adaptation of the System Usability Scale (Sharfina & Santoso, 2016). One negative item of the standard ten-item SUS was not included in the distributed form, so the score was rescaled to the 0–100 range with Equation (1), where R_i is the response to item

$$SUS = [\sum_{i \in \{1,2,4,6,8\}} (R_i - 1) + \sum_{j \in \{3,5,7,9\}} (5 - R_j)] \times 100/36 \quad (1)$$

Respondents were predominantly university students (86.7%), mostly aged 20–22 years (80.0%; mean age 21.9 years), and almost all were frequent smartphone users (96.7%), which matches the digitally literate target users but means that the scores are likely an upper bound for the general population. The descriptive statistics are given in Table 11. The mean SUS score of 71.6 lies above the commonly used average of 68, which places the application in the “acceptable” range with the adjective rating “good” (Bangor *et al.*, 2008, 2009) and at approximately the 61st percentile of the normative distribution (Lewis & Sauro, 2018; Lewis, 2018). Because the lower bound of the 95% confidence interval (66.1) is still slightly below 68, the claim of above-average usability is indicative rather than statistically established at this sample size. Internal consistency is acceptable (Cronbach’s $\alpha = 0.74$), comparable to reliability reported for SUS on Indonesian smartphone applications (Usman & Gustalika, 2022). Seventeen respondents (56.7%) scored above 70, twelve (40.0%) fell in the marginal band of 50–70, and one (3.3%) scored below 50. Three respondents gave the maximum response to every item, including logically opposed positive and negative items; excluding these straight-lined

responses raises the mean to 73.4 (95% CI [67.7, 79.0]) and α to 0.81, so the conclusion is robust to anomalous responses.

Table 11. Descriptive statistics of SUS scores for the citizen-reporter role (n = 30, nine-item adaptation).

Statistic	Value
Mean (SD)	71.6 (14.6)
Median	72.2
Minimum – maximum	33.3 – 97.2
95% confidence interval of the mean (t-based)	[66.1, 77.0]
Cronbach's α (nine items)	0.74
Acceptability (Bangor et al., 2009) / adjective rating (Bangor et al., 2008)	Acceptable / Good
Percentile rank (Lewis & Sauro, 2018)	$\approx$ 61st
Sensitivity: excluding three straight-lined respondents (n = 27)	73.4 [67.7, 79.0]; $\alpha = 0.81$

Score computed with Equation (1); CI = confidence interval.

The item-level contributions in Figure 7 show where the remaining usability problems lie. The highest contributions come from positive items on functional reliability and ease of use (P4, 3.47; P2, 3.33; P6, 3.27), whereas the three lowest come from negative items related to learnability and consistency: needing help from another person or a technician (P3, 1.90), needing to get used to the application first (P9, 2.20), and perceived inconsistency of the interface (P5, 2.27). Open-ended feedback was consistent with this picture: respondents valued the ease of reporting and the automatic agency suggestion and asked mainly for clearer feedback after a report is submitted. The usability barrier is therefore the initial learning curve and interface consistency, not core functionality, which points to onboarding guidance, consistent components and terminology, and post-submission status tracking as the next design iteration.

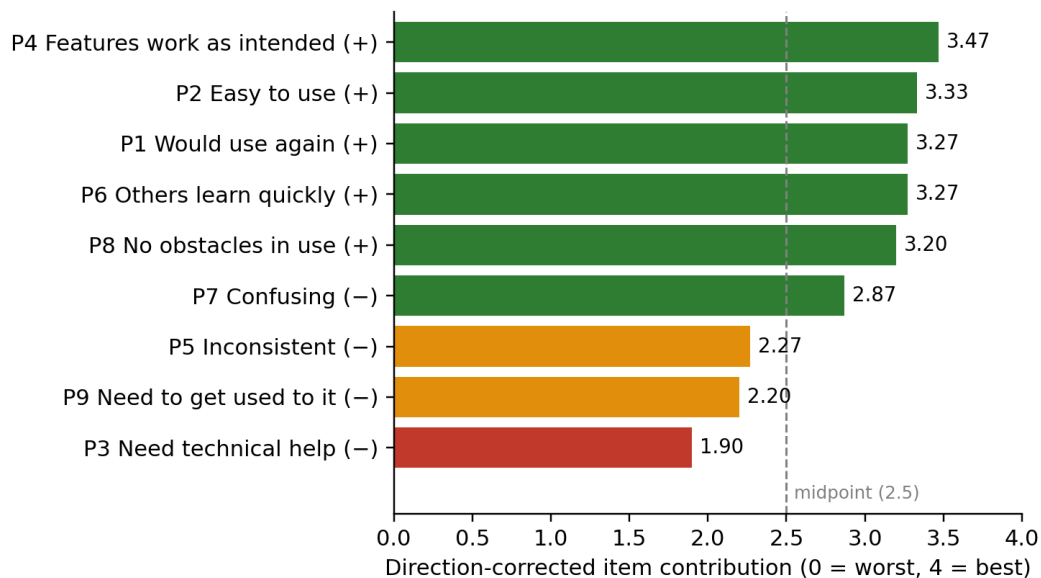


Figure 7. Direction-corrected contribution of each SUS item (0 = worst, 4 = best), ordered from strongest to weakest; items below the midpoint of 2.5 indicate priorities for improvement.

3.8 Error Analysis and Sources of Ambiguity

The row-normalised confusion matrix in Figure 8 reveals three dominant error patterns. First, Kelurahan has the lowest recall among the large classes (0.69); its reports are dispersed mainly to Dinas Perhubungan (134 samples), Dinas Sumber Daya Air (110), Other agencies (92), and Satpol PP (86), which is expected because neighbourhood-level complaints overlap the mandates of many agencies. Second, the residual Other agencies class is the hardest to separate (322 of 663 samples correct), because it contains heterogeneous reports that resemble other classes. Third, Dinas Sumber Daya Air suffers from low precision, because flooding and drainage reports that belong to Bina Marga or Kelurahan are predicted as water resources; 11% of true Water Resources reports are conversely predicted as Bina Marga. Contrary to the expectation that minority classes would be weakest, Dinas Cipta Karya, Tata Ruang, dan Pertanahan reaches a recall of 0.88.



Figure 8. Row-normalised confusion matrix of the deployed configuration on the test set (diagonal = per-class recall). Agency labels are abbreviated.

Three sources of ambiguity underline these patterns. The first is label noise introduced by mapping 487 raw agency names onto nine classes, since some reports have historically been assigned to different agencies depending on operator practice; label errors of this kind are known to distort both training and the apparent ranking of models on benchmark test sets (Northcutt *et al.*, 2021). The second is very short narratives without problem-specific keywords, such as “mohon ditangani” (“please handle this”), which weaken the text modality. The third is low-quality images, such as blurred or poorly lit photographs, which weaken the visual modality. All three are aleatoric, which is consistent with the conclusion in Section 3.3 that the performance ceiling is set by the data rather than by the model’s epistemic uncertainty. Operationally, this suggests treating the residual class and low-confidence predictions as a human-review queue rather than forcing an automatic assignment, and targeting the Kelurahan–Water Resources–Bina Marga boundary with clearer labelling rules.

3.9 Comparison with Related Work and Practical Implications

The closest predecessor, Hanif *et al.* (2026), reported a weighted F1 of 80.73% for a frozen DINOv2 + multilingual-E5 + CatBoost pipeline on 53,885 reports and eight agencies under cross-validation. The present configuration attains a weighted F1 of 0.8061 on a single held-out test partition of a larger dataset (61,773 pairs) with nine classes, including a difficult residual class. The two figures are not directly comparable because of the different protocols and label spaces, but they indicate that the

performance level of this architecture family is maintained under a stricter held-out evaluation and a broader label space. Text-only approaches on other Indonesian platforms (Zahro *et al.*, 2025) and image-only municipal systems (Shama *et al.*, 2024; Vrabie, 2025) address one modality; the ablation in Table 3 quantifies what each modality contributes when both are available. The design choice of a gradient-boosted head on frozen embeddings is also consistent with evidence that tree ensembles remain highly competitive on fixed-length tabular feature vectors (Grinsztajn *et al.*, 2022; Schwartz-Ziv & Armon, 2022), and in this study it made an exhaustive 16-pair encoder ablation feasible within one experimental session.

To make the position of this study explicit, Table 12 summarises prior studies that automate the classification or routing of citizen reports and urban events, and Table 13 contrasts the proposed pipeline with those approaches' aspect by aspect, together with the evidence reported in this article for each advantage. The advantages lie mainly in the breadth and rigour of the evidence and in deployability rather than in raw accuracy: on the only directly related dataset lineage, weighted F1 is on par with the predecessor (0.8061 versus 80.73%), but it is obtained under a stricter held-out protocol, a broader nine-class label space, an evidence-based encoder choice, and a validated serving path.

Table 12. Prior studies on automated classification and routing of citizen reports and urban events, compared with this study.

Study	Data and setting	Modality and method	Evaluation scope / reported result	Gap addressed by this study
Shama <i>et al.</i> (2024)	Municipal complaint reports, city corporation (Bangladesh); English; non-public data	Image only; classifier built with Teachable Machine; mobile task-distribution app	Image-classification model inside a mobile app; no encoder benchmarking or uncertainty reported	Narrative text is not used; no comparison of modern pre-trained encoders; no export-parity or uncertainty analysis
Vrabie (2025)	Municipal e-government platform (Romania); non-public data	Image analysis with conventional computer vision; text handled by a separate model	Image-level analysis for municipal responsiveness	Image and text are not fused into one decision; no uncertainty or serving validation
Zahro <i>et al.</i> (2025)	LaporGub Jawa Tengah complaint platform; Indonesian	Text only; XLM-RoBERTa with focal loss for skewed categories	Text-category classification under a skewed class distribution	Photographic evidence is ignored; no deployed routing pipeline
Tripathi <i>et al.</i> (2026)	Civic-issue management system (India); English	Image + text; ViT + BERT joint embeddings via CLIP	Multimodal civic-issue categorisation	No benchmark against frozen self-supervised encoders; English only; no uncertainty or export validation
Shen and Hu (2025)	Urban-governance event identification; general domain	Large multimodal models	Higher event-identification accuracy than manual categorisation; reduced operator workload	General-purpose, compute-heavy models; not evaluated for agency routing in a low-resource language
Hanif <i>et al.</i> (2026) — closest predecessor	Jakarta CRM; 53,885 reports; 8 agencies; Indonesian	Image + text early fusion; frozen DINOv2 + multilingual-E5; CatBoost with PGS	Weighted F1 = 80.73% under cross-validation; model level only	Two encoder choices per modality; PGS taken at face value; no ONNX export, deployment, routing, or usability evaluation
This study	Jakarta CRM; 61,773 image-text pairs; 487 raw labels → 9 agencies incl. a residual class	Image + text early fusion of L2-normalised frozen DINOv3-L + mE5-L (2048-d); CatBoost(+PGS); 4 × 4 encoder ablation	Held-out test (n = 9,266): macro-F1 0.7747, weighted F1 0.8061, balanced accuracy 0.7916, ROC-AUC 0.9667	ONNX parity, FastAPI/Flutter deployment, haversine routing, and SUS usability evaluated in the same study

Entries for prior studies summarise the settings and methods as described in the cited works; metrics obtained on different datasets, label spaces, and protocols are not directly comparable. PGS = Posterior Gaussian Sampling; DINOv3-L = DINOv3-Large; mE5-L = multilingual-E5-Large.

Table 13. Advantages of the proposed early-fusion pipeline over prior approaches, with the supporting evidence reported in this article.

Aspect	Prior approaches (Table 12)	Proposed approach	Advantage and evidence in this article
Use of modalities	Single modality (image only or text only), or CLIP-style joint embedding	Early fusion of an image embedding and an Indonesian narrative embedding	Fusion adds +2.32 pp macro-F1 over text only and +17.74 pp over image only (Table 3)
Encoder selection	Fixed by convention; the predecessor tested one pairing (DINOv2 + mE5)	Full 4 × 4 ablation over 16 image–text encoder pairs with an identical head	Choice is evidence-based: mE5-L is best in every row; the text encoder matters about four times more than the image encoder (4.07 vs. 0.95 pp) (Figure 4)
Language handling	English-language data, or Indonesian text only	Frozen multilingual encoder applied to informal Indonesian narratives	mE5-L outperforms the Indonesian-specific IndoBERT-Large-p2 and Cendol-mT5-Large (Figure 4)
Data and label space	53,885 reports / 8 agencies in the predecessor; non-public data elsewhere	61,773 authentic pairs; 487 raw agency labels consolidated into 9 classes, including a residual class	Larger and more realistic label space, with the real class imbalance retained (Section 2.1)
Evaluation protocol	Cross-validation with weighted F1 (predecessor)	Held-out stratified test set never used for training or selection; macro-F1, balanced accuracy, ROC-AUC, bootstrap 95% CI	Weighted F1 0.8061, on par with 80.73% but under a stricter protocol; macro-F1 95% CI [0.764, 0.784] (Section 3.1)
Class imbalance	Focal loss, or not addressed	Class-weighted CatBoost on the real distribution (no resampling)	Balanced accuracy 0.7916; minority class Cipta Karya/Tata Ruang reaches F1 = 0.8584 with 221 samples (Table 2)
Uncertainty	None, or PGS adopted without a controlled test	Confound-controlled PGS re-evaluation at fixed tree count with a 2,000-resample bootstrap; epistemic uncertainty as mutual information	Shows the naive +0.63 pp gain is confounded with tree count (its paired replication counterpart is not significant, Table 6) and PGS costs 0.46–0.84 pp at fixed trees; PGS is repositioned as a low-cost uncertainty signal (≈ 0.05 ms per sample) (Table 4); in a replication its epistemic signal is a weaker error detector than the maximum class probability (Table 7)
Training cost	End-to-end fine-tuning or large multimodal models	Frozen encoders with cached embeddings and a gradient-boosted head	308–602 s per head configuration, which made the 16-pair ablation feasible (Section 3.1)
Export and serving	Not reported	ONNX (opset 17) export with numerical-parity validation	$ \Delta p _{\max} = 1.38 \times 10^{-6}$ and 9,266/9,266 identical top-1 predictions; train–serving gap quantified (130 samples) (Table 8)
Deployment and routing	Model-level evaluation, or single-modality app	FastAPI server + Flutter/Supabase Android client with haversine nearest-office routing	45/45 routes resolved; about 1.08 s end-to-end on CPU; 216 MB client without on-device model (Tables 9–10)
User validation and oversight	Categories selected by users or officers (JAKI, LAPOR, Qlue); usability of the model pipeline rarely assessed	Predicted agency, confidence, and alternatives shown to the citizen, with manual correction before submission	SUS = 71.6 (acceptable; $n = 30$; Cronbach's $\alpha = 0.74$) (Table 11)

pp = percentage points; CI = confidence interval; SUS = System Usability Scale. Advantages refer to the scope and rigour of the evidence and to deployability; raw accuracy is on par with, not above, the closest predecessor.

For public-sector adoption, systematic reviews stress that AI in public administration must remain transparent, contestable, and subject to human oversight (Zuiderwijk *et al.*, 2021; Madan & Ashok, 2023). The prototype addresses this by showing citizens the predicted agency, its confidence, and alternative predictions, and by allowing manual correction before submission; these corrections can also serve as a stream of new labels for periodic retraining. Four practical lessons emerge for similar civic systems: (i) the choice of text encoder deserves more attention than the choice of image encoder

in narrative-rich complaint channels; (ii) uncertainty mechanisms should be evaluated with the training budget held fixed and validated along the full serving path; (iii) routing quality depends as much on the completeness of the office registry as on the classifier; and (iv) the plain maximum class probability of the argmax (served) inference path is the baseline against which any ensemble-based uncertainty signal should be judged, because in the replication run (Section 3.4) it was a stronger deferral signal than any PGS-based uncertainty score.

4. Conclusion

This study developed and evaluated an early-fusion multimodal pipeline that routes Indonesian citizen environmental reports to nine target agencies, together with a deployed client–server prototype. On an authentic dataset of 61,773 image–text pairs, frozen DINOv3-Large and multilingual-E5-Large embeddings classified by a CatBoost head achieved a test macro-F1 of 0.7747 (95% CI [0.764, 0.784]), a balanced accuracy of 0.7916, and a macro-ROC-AUC of 0.9667. Fusion improved macro-F1 by 2.32 pp over the strongest single modality (accuracy +2.00 pp; worst-case McNemar bound on top-1 correctness, $p \leq 0.003$), and a 16-pair encoder ablation showed that the text encoder is the decisive design choice, with multilingual-E5-Large best in every pairing.

Methodologically, the study showed that the apparent accuracy benefit of Posterior Gaussian Sampling disappears once the number of boosting iterations is held constant: at a fixed tree count, PGS inference lowers macro-F1 by 0.46–0.84 pp and accuracy by 0.43–0.82 pp (on the deployed checkpoint, the accuracy loss is significant even at the least favourable split of disagreements: worst-case McNemar $p \leq 7.6 \times 10^{-4}$), and the naive +0.63 pp difference, which is confounded with the tree count, could not be tested on paired samples in the original run; in a replication on a later snapshot of the same portal (37,503 pairs; test $n = 5,626$), its counterpart (+0.27 pp macro-F1, +0.39 pp accuracy) was not statistically significant (McNemar $p = 0.168$). PGS does not improve accuracy in this setting, because the dominant uncertainty in citizen reports is aleatoric. The same replication confirmed that PGS averaging lowers accuracy at a fixed checkpoint and showed that it worsens calibration (ECE 0.0379 versus 0.0271 under ordinary argmax inference, the mode served in production). In that replication, the maximum class probability of the argmax path flagged misrouted reports better than the PGS epistemic signal (AUROC 0.8041 versus 0.6447), and deferring the 20% least certain reports to operators would raise the accuracy of the automatically routed reports from 0.8102 (all reports routed) to 0.8809 (the 80% most certain routed). From an engineering perspective, the classifier head was exported to ONNX with numerical parity ($|\Delta p|_{\max} = 1.38 \times 10^{-6}$), but PGS does not survive the export, so the server model differs from the reported one on 130 of 9,266 test samples and lacks the PGS uncertainty terms; documenting and resolving such train–serving gaps is necessary for trustworthy deployment. The complete system routed all 45 verification cases to an office, ran end to end in about 1.08 s on CPU, kept the mobile package free of models, and achieved an acceptable SUS score of 71.6.

Future work should complete encoder-level parity verification, restore the PGS uncertainty terms in the serving path if they prove informative beyond the maximum class probability (Section 3.4), evaluate cross-regional generalization with data from other Indonesian complaint platforms, extend the selective-prediction analysis of Section 3.4 with conformal prediction for coverage-guaranteed deferral to human operators, and measure availability and cost for production-grade deployment. These steps would move the prototype from a

validated proof of concept towards an operational triage aid for smart-city complaint channels.

NOVELTY AND RESEARCH CONTRIBUTION

To meet the standard expected by a Q1 Scopus-indexed journal, this section states the novelty claims explicitly and at three levels — architectural, methodological, and engineering — positioned against the closest related work. Every claim below is grounded in an actual finding from the dissertation or, where marked, from the replication run of Section 3.4; none is invented for the purpose of this framing.

5.1 Primary Novelty Claims

1. This study builds upon the research of Hanif et al. (2026)—which previously demonstrated that an **early-fusion** workflow utilizing DINOv2, multilingual-E5, and CatBoost (incorporating PGS), alongside frozen visual components, achieved superior performance on this task (weighted-F1 score of 80.73%; 53,885 reports; 8 agencies). We extend this research across four aspects: upgrading the visual backbone to DINOv3 and justifying this choice through a comprehensive 4×4 encoder ablation analysis; scaling up the dataset to 61,773 image-text pairs covering nine agencies; re-evaluating the previously employed PGS mechanism by implementing confound control; and carrying the selected model through ONNX export validation, a production FastAPI/Flutter deployment, and real-user usability testing.
2. The primary scientific contribution is a re-evaluation of PGS while controlling for confounding factors. A simple comparison between the "with PGS" and "without PGS" conditions shows a +0.63-percentage point increase in the **macro-F1** score. However, the two variants utilize different checkpoints, with the PGS checkpoint employing nearly twice as many boosting trees (1,500 versus ≈2,989). We evaluated a fixed checkpoint under both inference modes and tested the difference with McNemar's exact test (worst-case bound $p \leq 7.6 \times 10^{-4}$ for the accuracy loss under PGS averaging), because the 2,000-resample bootstrap intervals are not a paired test (Section 3.3). Once this variable was controlled for, the accuracy improvement disappeared. PGS is therefore not a source of improved accuracy, and a replication with per-sample predictions (Section 3.4) shows that its epistemic-uncertainty signal is also weaker than the plain maximum class probability for flagging misrouted reports.
3. An engineering perspective, the CatBoost PGS virtual ensemble lacks an ONNX operator, meaning this component is omitted during the export process. Consequently, the model used in production differs from the one reported as the primary result: 130 out of 9,266 test samples yielded different top-1 predictions, with the production pipeline recording a higher score (accuracy of 0.8116 versus 0.8074). Such discrepancies between training and deployment (train-serving skew) are rarely quantified in civil engineering research. The validation covered equivalence between the ONNX and native formats ($|\Delta p|_{\max} = 1.38 \times 10^{-6}$; identical top-1 predictions across all 9,266 samples), end-to-end latency (1.08 seconds), Haversine-based routing across Jakarta's five administrative regions, and real-world usability (SUS score = 71.6, $n = 30$).

4. **Encoder selection grounded in evidence, not assumption.** Rather than committing to one encoder pair by convention, this study runs a full 4×4 ablation matrix (16 combinations) and documents that DINOv3-Large and EVA-02-Large perform near-identically while Hiera-Large is consistently the weakest across every text-encoder pairing — a design-space mapping rarely reported this completely in the citizen-reporting literature.
5. **Methodological contribution: a confound-controlled re-evaluation of an ensemble-based uncertainty method.** This is the most scientifically valuable contribution of the manuscript. The study identifies that the naive “with PGS” vs. “without PGS” comparison commonly reported in the uncertainty-quantification literature is confounded here with a near-doubling of boosting iterations (1,500 vs. $\approx 2,989$ trees). By evaluating one fixed checkpoint under two inference modes — rather than two separately trained checkpoints — and testing significance with exact McNemar tests (a worst-case bound in the original run and per-sample tests in the replication of Section 3.4) rather than with overlapping bootstrap intervals, the study shows that the commonly claimed “PGS improves accuracy” effect is not supported once the confound is controlled for. This directly engages with the ongoing discourse on reproducibility and confound mis-attribution in ensemble-based uncertainty quantification (cf. Hüllermeier and Waegeman, 2021, on the aleatoric/epistemic distinction that explains why averaging the virtual-ensemble members yields no accuracy gain here).
6. **Engineering contribution: exposing a production-serving gap rarely quantified in this literature.** The study shows empirically that CatBoost's PGS virtual-ensemble mechanism has no ONNX operator equivalent and is therefore silently dropped at export time. As a result, the model actually served in production (ONNX, no PGS) differs from the model reported as the headline result (native, with PGS) — 130 of 9,266 test samples receive a different top-1 prediction, and the production path in fact scores higher (accuracy 0.8116 vs. 0.8074). Train-serving skew of this kind is a well-known general ML-engineering failure mode, but it is rarely quantified this explicitly in the citizen-reporting/civic-technology literature, making this documentation itself a useful contribution for an MLOps- and smart-city-systems audience.
7. **End-to-end system validation rarely shown together.** Many multimodal classification studies stop at the model level (test-set metrics). This study complements the accuracy claim with (a) ONNX-vs-native numerical parity validation on the classifier head ($|\Delta p|_{\max} = 1.38 \times 10^{-6}$, 9,266/9,266 identical top-1 predictions); (b) per-stage latency benchmarking of the deployed pipeline (1.08 s end-to-end, with the encoders accounting for more than 99% of that time); (c) verification of the haversine-distance routing mechanism across 45 test points spanning Jakarta's five administrative cities; and (d) a real-user usability evaluation with the System Usability Scale (SUS = 71.6, $n = 30$, Cronbach's $\alpha = 0.74$), including a robustness check against straight-lining responses. The combination of accuracy, export parity, production latency, and real-user usability is rarely presented together in one civic-technology manuscript.
8. **Dataset contribution.** The 61,773-pair authentic, web-scraped Jakarta CRM dataset, with a documented mapping from 487 raw agency (SKPD) labels to nine target classes, is a contribution in its own right to Indonesian-language NLP and computer-vision research in the smart-city domain — a domain historically dominated by English-language datasets (Praveen et al., 2023) or by non-public datasets in comparable studies (Vrabie, 2025; Shama et al., 2024).

5.2 Position Relative to Related Work

Table A1 maps this study's position against six comparable studies in AI-based citizen-report or urban-event classification, ordered to make the incremental contribution over the closest predecessor (Hanif et al., 2026) explicit. This table is intended to anchor the manuscript's Related Work section.

Study	Modality	Encoder	Uncertainty?	Export & serving verified?	Language / domain
Shama et al. (2024)	Image only	Teachable Machine	No	No	English (Bangladesh)
Vrabie (2025)	Image (text separate)	Conventional CV	No	No	Romanian
Zahro et al. (2025)	Text only	XLM-RoBERTa + focal loss	No	No	Indonesian
Tripathi et al. (2026)	Image+text	ViT+BERT via CLIP	No	No	English (India)
Shen and Hu (2025)	Multimodal (general)	Large multimodal model	Not reported	No	General (urban governance)
Hanif et al. (2026) — closest predecessor	Image+text	DINOv2 + mE5 (frozen) + CatBoost+PGS	PGS used, not critically analyzed	No	Indonesian
This study	Image+text	DINOv3-L + mE5-L (frozen) + CatBoost, 4x4 ablation	PGS, confound-controlled + bootstrap test	Yes (ONNX parity + latency)	Indonesian

Table A1. Position of this study relative to six comparable AI-based citizen-report / urban-event classification studies, with Hanif et al. (2026) highlighted as the closest predecessor on the same dataset lineage.

6. SUGGESTIONS TO STRENGTHEN THE MANUSCRIPT FOR Q1 SUBMISSION (IJOSt)

This section summarizes what is already strong in the manuscript (and should be preserved), followed by what should be added — further analysis of data already collected in the dissertation, additional experiments, and supporting literature — to improve the manuscript's competitiveness for a Q1 Scopus journal such as IJOSt.

6.1 Already Strong — Preserve These

- The full 4×4 encoder ablation matrix (16 combinations) — rarely carried out this completely in comparable theses/papers; make it a headline result in Results, not a buried table.
- The confound-controlled PGS re-evaluation with bootstrap testing — this is the single most “Q1-worthy” element of the manuscript; give it proportional space in the Discussion, framed as a contribution rather than a limitation.
- The scientific transparency around the train-serving gap (PGS not surviving ONNX export) — Q1 reviewers generally reward this kind of honesty; frame it explicitly as Contribution 4 in Section 5.1, not merely a caveat.
- The full usability evaluation, including the robustness check against straight-lining respondents and the Cronbach's alpha reliability check — keep every methodological detail.

6.2 Additional Analyses Obtainable from Existing Data

- Pairwise statistical significance testing among the top encoder pairs in Table 4.8 (not only the PGS on/off comparison) — e.g., McNemar's test or paired bootstrap between DINOv3-L, EVA-02-L, and DINOv2-L — to convert the current “near-identical” claim into a formally tested one. [Addressed in Section 3.4 (replication run, later data snapshot); no pairwise difference is significant after Holm adjustment, which does not establish that the encoders are equivalent.]
- A reliability diagram and Expected Calibration Error (ECE) comparing CatBoost with and without PGS, to complement the “PGS's value lies in uncertainty, not accuracy” claim with quantitative calibration evidence rather than accuracy metrics alone. [Addressed in Section 3.4 (replication run); the result does not support this claim: PGS averaging worsens calibration (ECE 0.0379 vs. 0.0271 under ordinary argmax).]
- A correlation analysis between PGS epistemic uncertainty and prediction correctness (e.g., using uncertainty as a separator between correct and incorrect predictions via AUROC) — this would strengthen the practical argument that the uncertainty signal remains useful for triggering manual correction even though it does not raise accuracy. [Addressed in Section 3.4 (replication run); the result does not support this argument: the epistemic signal separates errors only weakly (AUROC 0.6447) and is outperformed by the maximum class probability (AUROC 0.8041).]
- A computational-cost comparison (parameter count / FLOPs) across the 16 encoder combinations in Table 4.8, paired with macro-F1, to present the accuracy-efficiency trade-off that Q1 reviewers in applied deep learning routinely expect.
- Additional quantitative error analysis: narrative length (token count) vs. prediction correctness, and a simple image-quality proxy (e.g., Laplacian variance as a sharpness

measure) vs. prediction correctness — to strengthen Section 4.11.2 (sources of ambiguity) with measured evidence rather than anecdotal examples.

6.3 Additional Experiments Worth Considering (Time/Resources Permitting)

- A fully fine-tuned end-to-end baseline (e.g., lightly fine-tuning CLIP or a ViT+IndoBERT pair end-to-end, rather than frozen encoders + CatBoost) to show the trade-off between the proposed frozen-encoder approach (computationally cheap, fast to reproduce) and full fine-tuning (potentially more accurate, far more expensive) — Q1 reviewers in applied machine learning almost always ask for this comparison.
- A cross-region generalization test by adding a small out-of-distribution evaluation set from the LaporGub Jawa Tengah platform (Zahro et al., 2025) — even at small scale, this would directly answer the geographic-coverage limitation already acknowledged in Section 4.11.3 of the dissertation while strengthening the generalizability claim.
- Throughput and latency measurement under GPU acceleration (not CPU only) to complement Table 4.17, since systems/MLOps reviewers are likely to ask for production-realistic numbers rather than development-environment CPU figures alone.
- A sensitivity analysis of a selective-prediction (reject-option) threshold built on the PGS epistemic-uncertainty signal — this complements the conformal-prediction direction already suggested in Section 4.6.2 of the dissertation as future work, and is a relatively inexpensive additional experiment since it only requires re-analyzing predictions already produced. [Addressed in Section 3.4 on predictions of a new replication run; a threshold on the epistemic signal is the weakest deferral rule (coverage at 90% accuracy 0.424 vs. 0.724 for the maximum class probability).]

6.4 Gaps That Should Be Resolved or Explicitly Framed Before Submission

- Numerical-parity verification of the image encoder (DINOv3-Large) ONNX export is not yet complete in the dissertation (Table 4.9) — completing this before submission is recommended, since reviewers will likely question the currently partial answer to RQ3.
- Availability and operating cost under RQ5 are not empirically measured in the dissertation — consider adding a simple uptime/cold-start measurement, or at minimum framing both explicitly in the manuscript as future work, so RQ5 does not read as fully answered when it is only partially answered.
- Ensure IJoST's official Author Guidelines and Article Template (<https://ijost.upi.edu>) are followed consistently, including citation style, page limits, and the structured-abstract format (objective, methods, results, discussion, impact) that IJoST requires.

6.5 Additional Post-2020 Supporting Literature for the Related Work Section

The following sources, identified through a targeted literature search, are recommended to strengthen the Related Work / Discussion sections. All are dated 2021 or later, consistent with the requirement to keep supporting and comparative citations current; foundational method and instrument citations (CatBoost, DSRM, SUS, PGS) are retained at their original publication year in the main reference list because they describe the exact methods and instruments used in this study, not comparative literature.

- Hanif, I. A., Tjitrahardja, E., Naufal, R. B., & Rahadiani, L. (2026). *Automated assignment of community reports with early fusion multimodal transformer*. Jurnal Ilmu Komputer dan Informasi, 19(1), 1–9. — the closest predecessor study; strongly recommended as the anchor citation for the Related Work section.
- Li, S., & Tang, H. (2025). *Multimodal alignment and fusion: A survey*. IEEE Transactions on Pattern Analysis and Machine Intelligence (preprint arXiv:2411.17040).
- Hüllermeier, E., & Waegeman, W. (2021). *Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods*. Machine Learning, 110(3), 457–506.
- Praveen, S. P., et al. (2023). *Effective disaster management through transformer-based multimodal tweet classification*. Revue d'Intelligence Artificielle, 37(5), 1263–1272.
- Shama, F., Aziz, A., & Deya, L. B. M. (2024). *CitySolution: A complaining task distributive mobile application for smart city corporation using deep learning*. SoftwareX, 27, 101829.
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ACKNOWLEDGED LIMITATIONS

Consistent with the underlying dissertation, the manuscript should retain an explicit limitations paragraph: (i) ONNX numerical-parity verification is complete for the classifier head but not yet independently re-verified for the image encoder; (ii) the PGS ablation's headline comparison conflates PGS inference with iteration count, a confound this study explicitly resolves rather than hides; (iii) usability evaluation covers only the citizen-reporter role and used a nine-item (not standard ten-item) SUS instrument on a sample skewed toward digitally literate students; (iv) the dataset's geographic coverage is limited to DKI Jakarta, so cross-region generalization is untested; (v) availability and operating-cost consequences of the on-server inference decision (RQ5) are discussed qualitatively but not empirically measured in this study; and (vi) the calibration, paired-test, and selective-prediction results in Section 3.4 come from a replication run on a later mirror of the same portal, with a timm port of the DINOv3 weights, rather than from the original checkpoints, and they capture test-sample variability for a single training seed per configuration, not variability across training seeds (Bouthillier et al., 2021).

